

MagicCode Super-Resolution Software API

Reference

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Header: interface/mc_interface.h. Enable helper: interface/mc_enable.h.

1. Overview

The native API is a C interface for initialization, processing, control, status query, release, and version query. C++ projects include the header normally; symbols use extern "C".

2. Core Workflow

1. Zero-init input_param_t and set struct_size, input_type, width, height, scaler_factor, alg_mode, backend, and (for temporal) gpu_context plus create-stage depth/HDR.
2. Call MC_Init and keep the opaque handle.
3. Each frame, fill magic_frame_t (image_in / image_out). Spatial may leave frame NULL. Temporal must set frame to a temporal_frame_t with struct_size, depth, motion, jitter, and camera fields.
4. Call MC_Process(handle, &magic_frame).
5. Optionally call MC_Control with SET_PARAM or QUERY_STATUS.
6. Call MC_Uninit exactly once for each successfully initialized handle.

3. Enumerations

Enum	Values	Notes
cmd_params_e	SET_PARAM, QUERY_STATUS	Control command.
input_type_e	INPUT_BUFFER / R8, INPUT_TEXTURE_RGB8Unorm, INPUT_TEXTURE_R8Unorm, INPUT_BUFFER_RGB	CPU buffers vs GPU textures.
alg_mode_e	SPATIAL_SPEED_MODE, SPATIAL_BALANCED_MODE, TEMPORAL_SPEED_MODE, TEMPORAL_BALANCED_MODE	0..3; MAX_ALG_MODE=4.
magic_backend_e	DEFAULT, X86, NEON, METAL, OPENGL, OPENGLES, VULKAN, D3D11	Runtime selector.
mc_mv_jitter_e	UNSPECIFIED/EXCLUDED/ INCLUDED	On temporal_frame_t.mv_jitter.
mc_depth_reversed_e / infinite / hdr	UNSPECIFIED plus conventional/reversed, finite/infinite, LDR/HDR	Create-stage on input_param_t.

4. Structures

4.1 input_param_t

- struct_size: set to sizeof(input_param_t); 0 means current sizeof.
- width/height: [64, 4032]. scaler_factor spatial [1, 8], temporal (1, 8].
- alg_mode: four public modes listed above.
- gpu_context: copied at MC_Init; immutable. Vulkan temporal requires physical_device + device. D3D11 temporal requires device. Metal NULL device = system default.
- depth_reversed / depth_infinite / hdr_color: temporal create-stage; ignored for spatial.

4.2 magic_frame_t and temporal_frame_t

- magic_frame_t.image_in / image_out hold color and output magic_resource_t.
- magic_frame_t.frame is NULL for spatial and required for temporal.
- magic_frame_t.command_buffer: Vulkan temporal recording VkCommandBuffer; optional MTLCommandBuffer on Metal; unused on D3D11/GL/GLES.
- temporal_frame_t.struct_size must cover the v2 prefix through transparency. Tail fields (mv_jitter, exposure_texture) are read only when struct_size covers them.
- Required temporal resources: depth, motion. Optional: reactive, transparency, exposure_texture (1×1 R32F).

4.3 magic_resource_t

handle is a union: pointer (CPU / MTLTexture* / ID3D11Texture2D*), vk_image (VkImage), or gl_texture (GLuint). Exactly one member is live, chosen by the MC_Init backend. format is backend-native; layout is VkImageLayout (zero for other backends). No width/height: matching handle sizes is a caller contract.

4.4 control_param_t / output_status_params_t

SET_PARAM updates width, height, scale, mode, and model path. QUERY_STATUS returns dimensions, scale, mode, backend, gpu_time, error_code, and when temporal_status_valid is set, the last successful temporal snapshot (scalars only). GPU handles and command_buffer are never stored.

5. Functions

Function	Purpose	Return
void* MC_Init(input_param_t*)	Create handle	Opaque handle or NULL
int MC_Process(void*, magic_frame_t*)	One spatial or temporal frame	0 or negative MC_ERROR_*
int MC_Control(...)	SET_PARAM or QUERY_STATUS	0 or negative
int MC_Uninit(void*)	Release handle	0 or negative
char* MC_GetVersion(void)	Version string	Static "v2.0.0"

5.1 Vulkan temporal lifecycle

Keep the MC handle across frames. MC_Process(void*, magic_frame_t*) only records into the caller-owned recording VkCommandBuffer. Set image_in.layout, temporal depth/motion.layout, and image_out.layout to the real VkImageLayout at entry (0 is UNDEFINED and fails). The library does not begin, end, submit, or wait. After the caller submits and waits, call MC_Uninit at shutdown — not immediately after MC_Process while GPU work may still be in flight.

6. Enable helper (mc_enable.h)

MC_Enable / MC_Enable_3params / MC_Enable_4params lazily create a session from one GPU texture. Scale is [1.0, 8.0]. Mode range includes temporal enums, but Enable cannot pass depth/motion/jitter. Complete temporal integration must use the session API. Enable_4params accepts backends through MAGIC_BACKEND_VULKAN and does not accept MAGIC_BACKEND_D3D11. Windows Enable default is OpenGL.

Temporal Super-Resolution

v2.0.0 adds temporal super-resolution through the session API: libmagic_sr.a (Windows: libmagic_sr.lib) + mc_interface.h + MC_Init / MC_Process / MC_Uninit. MC_Enable remains a single-texture spatial helper. It may accept a temporal alg_mode_e value by range check, but it cannot supply per-frame depth, motion, or jitter and must not be documented as a complete temporal entry.

Modes

Value	Enumerator	Role
0	SPATIAL_SPEED_MODE	Spatial, throughput first
1	SPATIAL_BALANCED_MODE	Spatial, quality/cost trade-off
2	TEMPORAL_SPEED_MODE	Temporal, throughput first
3	TEMPORAL_BALANCED_MODE	Temporal, canonical FSR-family path

- Temporal scaler_factor is (1, 8]; 1.0 itself is rejected. Spatial remains [1.0, 8.0].
- TEMPORAL_SPEED_MODE on Metal, Vulkan, and OpenGLS currently uses a lightweight Convert → Activate → Upscale pipeline. Direct3D 11 and desktop OpenGL 4.3 Speed use the FSR-family temporal path (same family as Balanced, not a separate customer API).
- TEMPORAL_BALANCED_MODE uses the canonical FSR-family path on every GPU temporal backend. Desktop Balanced edge-resolve is supported on Windows Vulkan, Windows Direct3D 11, desktop OpenGL 4.3, and Apple Metal. Android Vulkan and Android OpenGLS do not run edge-resolve. If enable_sharpening is on, RCAS runs after edge-resolve.
- CPU backends (X86 / NEON) cannot run temporal modes.
- The API accepts (1, 8]. Exact 2× has optimized kernels. Not every scale or device is claimed as device-run.
- Internal pipeline names are implementation detail, not public configuration.

Minimum per-frame inputs

- `magic_frame_t.image_in / image_out`: caller-owned color and output GPU resources.
- `magic_frame_t.frame`: non-NULL `temporal_frame_t*` with `struct_size = sizeof(temporal_frame_t)`.
- `temporal_frame_t.depth` and motion at input resolution.
- `jitter_offset_x/y` in input pixels, top-left, +X right, +Y down.
- `frame_index / reset_history`; `camera_near / camera_far / camera_fov_y` (radians).
- `gpu_context` at `MC_Init`: Vulkan requires `physical_device + device`; D3D11 requires `device`; Metal device may be NULL; GL/GLES use the current context (library never makes current).
- Vulkan temporal requires `magic_frame_t.command_buffer` to be a recording `VkCommandBuffer`. Metal `command_buffer` is optional. D3D11 / GL / GLES leave it NULL.
- Create-stage `depth_reversed / depth_infinite / hdr_color` on `input_param_t` (all-zero = conventional-Z, finite far, LDR). Immutable for the handle lifetime.

Motion, depth, and masks

- Motion is current → previous. The library never negates MVs.
- `motion_vector_scale_x/y` is the only MV numeric conversion. (0,0) defaults to (input_width, input_height). Mixed zero is rejected.
- Depth is GPU device depth in [0, 1]. Linear view-space depth is not accepted.
- `reactive / transparency` are optional. Explicit caller textures win per channel.
- When omitted, the library derives an approximate mask. That is not a semantic transparency mask.
- Speed on Metal/Vulkan/GLES may derive both missing reactive and an approximate transparency. FSR-family paths (Balanced on all backends; Speed on D3D11/desktop GL) derive missing reactive only and bind an internal zero transparency texture.
- Both temporal modes enable depth+MV history rejection by default.
- Diagnostic only: `MAGICSR_TAAU_AUTO_MASK=0` and `MAGICSR_TAAU_HIST_REJECT=0` can turn those defaults off. They are not a customer configuration path.

7. Selected temporal error codes

- `MC_ERROR_TEMPORAL_NULL_FRAME` (-105001)
- `MC_ERROR_TEMPORAL_STRUCT_SIZE` (-105002)
- `MC_ERROR_TEMPORAL_MISSING_COLOR/OUTPUT/DEPTH/MOTION` (-105003..-105006)
- `MC_ERROR_TEMPORAL_NONFINITE / CAMERA_NEAR_FAR / MV_SCALE` (-105007, -105008, -105011)
- `MC_ERROR_TEMPORAL_FORMAT` (-105015); Vulkan UNDEFINED layout is rejected
- `MC_ERROR_TEMPORAL_COMMAND_BUFFER` (-105017) Vulkan recording buffer required
- `MC_WARNING_TEMPORAL_FRAME_INDEX` (105001) not last+1